

HARIS WAHEED BHATTI

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PROFILE

European Master's candidate in Building Information Modelling (BIM), specializing in Information Management in Construction, with hands-on experience in IFC data modeling, knowledge graphs, and BIM software development. Focused on applying AI, LLMs, and web-based BIM tooling to solve data-interoperability challenges in the AECO industry. Recipient of a PKR 100,000 grant from the Pakistan Engineering Council for a blockchain-based BIM transparency thesis project.

EDUCATION

European Master in Building Information Modelling <i>University of Ljubljana</i>	15/11/2025 – 30/09/2026 <i>Slovenia</i>
Field of study: Information Management in Construction – multidisciplinary program in BIM-integrated design, construction, and operation delivered jointly by leading European universities (bimplus.org). Thesis in progress.	
Master BIM Software Developer (Professional Training) <i>That Open Company</i>	01/12/2024 – Current <i>Online</i>
- Built browser-based BIM applications using TypeScript, HTML/CSS, Three.js, and WebIFC via the "That Open Engine" library - Applied these tools to manipulate, visualize, and query IFC data directly in the browser	
BE Civil Engineering <i>National University of Science and Technology</i>	28/08/2023 – 31/05/2024 <i>Pakistan</i>
Final grade: 3.25 Thesis: <i>Enhancing Construction Transparency and Dispute Resolution Through Blockchain-Based BIM Models</i> (see Projects) – bit.ly/49dUTmK	

PROJECTS

IFC Knowledge Graph & RAG Pipeline <i>Neo4j, LLMs, IFC</i>	30/01/2026
- Modeling building data as a property graph in Neo4j, preserving hierarchical and relational structures of BIM entities - Implementing RAG architecture combining LLMs and graph retrieval to enable semantic querying and knowledge discovery within BIM datasets - Building an automated pipeline for IFC ingestion and graph serialization, supporting bidirectional data flow between BIM files and graph environments	
Pakistani-Themed Expo BIM Management <i>IFC, Autodesk Construction Cloud</i>	28/10/2024 – 19/11/2024
- Spearheaded a BIM management project for a Pakistani-themed expo in Guimarães, Portugal, leveraging IFC and Autodesk Construction Cloud to develop a comprehensive technical design - Facilitated collaboration between Architecture and MEP disciplines, creating a federated model that integrated diverse design inputs - Conducted clash detection and compliance checks to ensure the model met all information requirements and design standards	
Blockchain-Based BIM Document Management <i>Solidity Bachelor's Thesis Project</i>	08/09/2023 – 05/04/2024
- Developed a smart contract using Solidity to digitally sign and manage construction documents, with off-chain storage linked to the blockchain for immutability and transparency - Funded by a PKR 100,000 grant from the Pakistan Engineering Council; designed to streamline dispute resolution for construction stakeholders - Link: sepolia.arbiscan.io/address/0x8535f1840d47b65ab11ac7d95b5430d017a4c867	

EXPERIENCE

Contract Management Intern <i>Engineering and Management Services</i>	07/08/2023 – 01/09/2023 <i>Pakistan</i>
Reviewed construction contract reports and arbitration case documentation under FIDIC conditions, and presented findings on FIDIC and arbitration procedures to the internal legal team	
Research Intern <i>Southern Illinois University Edwardsville</i>	17/07/2023 – 24/08/2023 <i>USA</i>
Extracted and synthesized findings from academic articles and abstracts to support a faculty-mentored research proposal, applying feedback to refine presentation and analysis technique	
Civil Engineering Intern <i>Fatima Fertilizer Company</i>	11/07/2023 – 04/08/2023 <i>Pakistan</i>
Modeled a 42m x 162m (6,804 m²) rock phosphate storage shed independently in 3D from 2D drawings, coordinating with stakeholders to refine the design	

SKILLS

Technical: IFC, TypeScript, Python, Artificial Intelligence, Knowledge Graphs, Claude / LLM tooling
Languages: English (C2), Urdu (C2)

CERTIFICATIONS

Coursera Certificates

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| Supervised Machine Learning: Regression and Classification | 19/02/2023 |
| <ul style="list-style-type: none">• DeepLearning.AI course covering Python-based AI algorithms, applied in the "AI for Built Environment" course. (coursera.org/verify/T9ZJ8HYJX2E7) | |
| Management of Urban Infrastructures – Part 1 | 15/02/2024 |
| <ul style="list-style-type: none">• EPFL course on how cities operate and how infrastructure is built, operated, maintained, and governed. (coursera.org/verify/MWW4ZWUXSTDP) | |
| Innovative Governance of Large Urban Systems | 21/02/2024 |
| <ul style="list-style-type: none">• Course on governing large urban cities while balancing infrastructure, policy, and citizen well-being. (coursera.org/verify/3VNHK3YRQ7MZ) | |